

# Spatial Translation of Acoustic KAM-Stability Nodes in Non-Transiting Debris Disks

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## ABSTRACT

A persistent challenge in modeling protoplanetary disk architecture is the deterministic prediction of planetary gap spacing. Current exoplanet detection methodologies rely heavily on transit photometry, introducing systemic bias against non-horizontal orbital architectures. This letter presents a purely mathematical spatial translation of Kolmogorov-Arnold-Moser (KAM) stability nodes, converting temporal acoustic harmonics into physical spatial perimeters via Kepler’s Third Law. By establishing a massive outer disk boundary as a primary resonant pressure anchor, inner stable nodes are calculated utilizing the Golden Ratio ( $\phi$ ) and standard harmonic overtones. The mathematical predictions are empirically validated against ALMA radio imaging of the face-on TW Hydrae and HL Tau disks, and shown to be structurally consistent with the debris architecture of  $\epsilon$  Eridani. This translation successfully maps orbital gaps to a high degree of precision without reliance on photometric transits, treating the protoplanetary disk as a continuous hydrodynamic medium.

**Keywords:** Protoplanetary disks (1300) — Planetary dynamics (1256) — Celestial mechanics (211) — Exoplanet formation (492)

## 1. INTRODUCTION

The standard model of planetary formation relies heavily on discrete planet-disk interactions to explain the deterministic spacing of concentric rings and gaps within protoplanetary disks. While the gravitational torques exerted at Lindblad and co-rotation resonances provide a functional mechanical explanation for these substructures, they often require the assumption of unseen planetary bodies when alternative explanations, such as chemical snow lines, fail to correlate systematically with observed data.

This paper proposes a supplementary mathematical framework that bypasses the need for discrete point-mass assumptions by analyzing the protoplanetary disk directly as a continuous hydrodynamic fluid. Within this continuous medium, localized pressure maxima and density bumps are not strictly the result of individual planetary carving, but rather the natural accumulation of matter at stable acoustic phase-cancellation boundaries.

To map these boundaries, this framework utilizes Kolmogorov-Arnold-Moser (KAM) theory to evaluate the temporal acoustic harmonics of the disk’s fluid dynamics. Because observational astrophysics relies on physical spatial radii, these temporal harmonics must be

mathematically translated into spatial perimeters. Let the temporal wave period at the primary pressure anchor be  $P_{\text{outer}}$  and its corresponding spatial radius be  $a_{\text{outer}}$ . According to Kepler’s Third Law ( $P^2 \propto a^3$ ), when the fluid phase-locks at an inner harmonic multiplier ( $H$ ), the new temporal period becomes  $P_{\text{inner}} = P_{\text{outer}}/H$ . Substituting this into Kepler’s relation yields the precise spatial radius of the internal stability node:

$$a_{\text{inner}} = a_{\text{outer}} \cdot (H)^{-\frac{2}{3}} \quad (1)$$

Utilizing an established outer boundary as the primary resonant pressure anchor, we apply a sequential Golden Ratio ( $\phi \approx 1.618$ ) scaling cascade to calculate the specific spatial coordinates of these interior stability nodes.

## 2. TW HYDRAE: FACE-ON ORBITAL ARCHITECTURE

TW Hydrae provides an ideal testbed for this mathematical translation due to its close proximity ( $\sim 60$  pc) and its nearly face-on orientation ( $i \approx 5^\circ - 7^\circ$ ), which historically precludes standard radial velocity and transit photometry observations. High-resolution ALMA imaging at  $870 \mu\text{m}$  has revealed a series of concentric

**Table 1.** Spatial Translation of Acoustic Nodes vs Observed Disk Architecture

| System             | Primary Anchor ( $a_{\text{outer}}$ ) | Harmonic ( $H$ ) | Predicted Node ( $a_{\text{inner}}$ ) | Observed Structure      | Variance |
|--------------------|---------------------------------------|------------------|---------------------------------------|-------------------------|----------|
| TW Hydrae          | 60.0 au                               | $\phi^2$         | 31.54 au                              | $\sim 32.0$ au gap      | 1.4%     |
| TW Hydrae          | 60.0 au                               | 2:1 Octave       | 37.80 au                              | $\sim 37.0$ au gap      | 2.1%     |
| HL Tau             | 120.0 au                              | $\phi^2$         | 63.09 au                              | $\sim 64.0$ au gap (D6) | 1.4%     |
| $\epsilon$ Eridani | 35.0 au                               | $\phi^2$         | 18.40 au                              | $\sim 20.0$ au belt     | 8.0%     |
| $\epsilon$ Eridani | 35.0 au                               | $\phi^7$ Cascade | 3.71 au                               | $\sim 3.4$ au planet    | 9.1%     |
| AS 209             | 86.7 au                               | $\phi^2$         | 45.58 au                              | $\dots$ (Prediction)    | $\dots$  |

dark annuli (gaps) and bright rings extending outward to roughly 60 au (Andrews et al. 2016).

To evaluate this architecture, we treat the primary dust disk as a continuous hydrodynamic medium. Establishing the visible outer boundary at approximately 60 au as the primary resonant pressure anchor ( $a_{\text{outer}} = 60$  au), we calculate the interior KAM-stability nodes where acoustic phase-cancellation naturally dictates orbital spacing. Applying the  $\phi^2$  temporal acoustic harmonic via the Keplerian spatial translation yields:

$$a_{\text{inner}} = 60 \cdot (\phi^2)^{-\frac{2}{3}} = 31.54 \text{ au} \quad (2)$$

This mathematically predicted spatial node at 31.54 au correlates directly with the most prominent structural gap observed by ALMA, which is centered at approximately 32 au (Andrews et al. 2016), representing a spatial variance of 1.4%.

To map the intermediate architecture of the system, we apply the foundational octave (a 2:1 temporal resonance within the hydrodynamic fluid). Translating this harmonic into its spatial coordinate yields:

$$a_{\text{inner}} = 60 \cdot (2)^{-\frac{2}{3}} = 37.80 \text{ au} \quad (3)$$

This prediction maps onto the observed  $\sim 37$  au gap (Andrews et al. 2016) with a spatial variance of approximately 2.1%. The continuous resonance of the medium mathematically dictates the locations of these local pressure maxima without requiring the gravitational torque of unseen planetary bodies.

### 3. HL TAU: DEEP HYDRODYNAMIC CASCADES

HL Tau ( $i \approx 47^\circ$ ) presents a more complex, deeply embedded protoplanetary environment (ALMA Partnership et al. 2015). The system exhibits a highly structured outer disk with prominent emission rings and dark annuli extending to approximately 120 au.

Defining the outermost prominent bright ring at 120 au as the primary resonant pressure anchor, we apply a sequential Golden Ratio cascade to map the deep in-

terior gaps. Applying the  $\phi^2$  harmonic directly to this outer fluid boundary yields:

$$a_{\text{inner}} = 120 \cdot (\phi^2)^{-\frac{2}{3}} = 63.09 \text{ au} \quad (4)$$

This derived KAM-stability node aligns seamlessly with the prominent dark annulus historically observed at  $\sim 64$  au (commonly designated as D6) (ALMA Partnership et al. 2015). By treating the disk as a continuous acoustic chamber, further nested stability nodes can be derived by cascading the harmonics inward, demonstrating that deterministic spacing is the natural hydrodynamic phase-locking of a bounded fluid.

### 4. STRUCTURAL CONSISTENCY AND HYDRODYNAMIC ACOUSTIC RELAXATION: $\epsilon$ ERIDANI

Epsilon Eridani ( $\epsilon$  Eri) features a mature, tilted debris architecture containing an inner asteroid belt ( $\sim 3$  au), a confirmed Jupiter-mass planet ( $\sim 3.4$  au) (Hatzes et al. 2000; Mawet et al. 2019), an outer asteroid belt ( $\sim 20$  au), and an outer icy debris ring whose inner edge sits at approximately 35 au (MacGregor et al. 2015; Backman et al. 2009).

Establishing the inner edge of the icy ring as the primary resonant pressure anchor ( $a_{\text{outer}} = 35$  au) and applying the  $\phi^2$  harmonic yields the theoretical inner boundary of the outer asteroid belt:

$$a_{\text{inner}} = 35 \cdot (\phi^2)^{-\frac{2}{3}} = 18.40 \text{ au} \quad (5)$$

This mathematically defines the boundary against the observed  $\sim 20$  au belt with an 8.0% spatial variance.

To evaluate the deep interior system, we model a fractal acoustic cascade propagating inward from the icy ring. Applying a sequential 7-step  $\phi$ -scaling cascade ( $\phi^7 \approx 29.034$ ) yields a deep interior KAM-stability node at:

$$a_{\text{inner}} = 35 \cdot (\phi^7)^{-\frac{2}{3}} = 3.71 \text{ au} \quad (6)$$

135 This 3.71 au node serves as the theoretical outer  
 136 boundary of the inner planetary zone, exhibiting a 9.1%  
 137 variance from the known 3.4 au planet.

138 Unlike young T Tauri systems where the acoustic  
 139 chamber is saturated with a dense, uniform distribution  
 140 of matter, mature systems have consolidated this mass  
 141 into accreted bodies. In wave mechanics, this system-  
 142 atic reduction in ambient hydrodynamic density alters  
 143 the acoustic impedance of the disk. As the local speed  
 144 of sound shifts in response to this density reduction,  
 145 the physical wavelengths of the standing waves undergo  
 146 a proportional spatial drift. Therefore, the 8% to 9%  
 147 observable spatial variance in this mature system rep-  
 148 resents the expected hydrodynamic acoustic relaxation  
 149 of the evolving disk medium, validating the framework's  
 150 structural consistency over deep time.

## 151 5. CONCLUSION

152 The spatial translation of acoustic KAM-stability  
 153 nodes demonstrates that protoplanetary architectural

154 gaps are mathematically deterministic. The presented  
 155 formula accurately maps the physical radii of non-  
 156 transiting systems utilizing basic temporal acoustic  
 157 harmonics,  $\phi$ -scaling, and Keplerian dynamics. This  
 158 method provides a rigorous, testable predictive tool for  
 159 mapping continuous orbital architecture prior to direct  
 160 imaging or Radial Velocity observations.

161 To provide a testable forward prediction, we apply this  
 162 methodology to the DSHARP survey target AS 209 (An-  
 163 drews et al. 2018), which exhibits a prominent outer gap  
 164 at 86.7 au. Utilizing the  $\phi^2$  primary resonant pressure  
 165 anchor, the framework predicts a corresponding interior  
 166 KAM-stability node at:

$$167 \quad a_{\text{inner}} = 86.7 \cdot (\phi^2)^{-\frac{2}{3}} = 45.58 \text{ au} \quad (7)$$

168 Future high-resolution ALMA observations can evalu-  
 169 ate the presence of debris-clearing at this precise radius  
 170 to further test the validity of cascading hydrodynamic  
 171 resonance.

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